Preregistration

Machine learning classification of stress intervention

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Study Information

Title	Machine learning classification of stress intervention
Description	Overview In Thayer & Stevens (2021), we conducted two experiments investigating the effects of human-animal interaction (HAI) on affect and cognition. In these experiments, participants experienced cognitive (working memory, attentional control) tasks before and after either a three-minute exposure to a dog (HAI condition) or a control task. Self-report measures of affect (mood, anxiety, stress) were collected repeatedly throughout the sessions. While interacting with a dog influenced measures of affect, they did not influence cognition. In addition to the affect and cognition measures, participants wore an Empatica E4 to record heart rate, electrodermal activity, body temperature, etc. However, we did not analyze these data, as the sampling rate was

too low for robust measures of heart rate variability (Malik et al. 1996, Laborde et al. 2017).

Arce & Gehringer (2021) have investigated machine learning algorithms that take Empatica E4 data and classify whether participants were experiencing stress induction. This offers the opportunity to apply the machine learning algorithms developed to classify stress to the Thayer & Stevens (2021) data and address two specific aims:

1. Assess the predictive accuracy of the machine learning algorithms in classifying intervention exposure
2. Generate an index of physiological stress from Empatica E4 output

Research questions

1. How accurately can machine learning algorithms classify stress intervention exposure?
2. Which algorithms best predict stress intervention exposure?
3. Does the accelerometer data act as a naive feature that gives extra weighting to the different affect states?
4. Which features best predict passive and active stress intervention exposures?

Hypotheses

1. Machine learning algorithms will classify psychological state at comparable or higher accuracy as observed in stress induction studies.

We expect that predictive accuracy for our stress intervention study will match or exceed those calculated by Arce and Gehringer (2021) for a stress induction study.

2. Binary classifiers will predict better than regression or multi-class classifiers.

Psychological state can either be binarized into cognitive load (combine pre-intervention cognitive load, post-active intervention cognitive load, and post-passive intervention cogntive load) and intervention states (combine active intervention and passive intervention) or include more categories (cognitive load, active intervention,

passive intervention, post-active intervention, post-passive intervention). Following Arce and Gehringer (2021), we expect the binary classifier to outperform other classifiers.

3. Accelerometer-based features will best predict binary states.

When the accelerometer features are included in the machine learning algorithm training, the movement associated with the active intervention will make them the most important features for classification. Excluding accelerometer features should improve the performance of the model and improve generalization. This is due to the active intervention involving an interaction between subject and a dog, so we expect that the accelerometer features would be more prominent when compared to the passive intervention due to actions like petting. While this would give the model an easy way to discriminate between the two interventions, it is not our intention to classify based on movement, but rather psychological state.

4. In terms of physiological markers, electrodermal activity-based features will best predict psychological state.

Given an appropriately trained classifier, the higher order features from the electrodermal activity signal should be the best features to predict passive and active stress intervention exposures. Considering that electrodermal activity is controlled by the sympathetic nervous system and is particularly sensitive to high arousal states (Schmidt et al., 2018), we would expect those features to correlate to an increase or decrease of cognitive load.

Design Plan

Study type	**Other**. Please explain.
Blinding	No blinding is involved in this study.

Study design Our study is going to utilize the Biosensor Framework, a C# library to process and evaluate machine learning algorithms for time-series data from a wrist worn physiological sensor. This library was validated against the Wearable Stress and Affect Detection (WESAD) dataset, which consists of 15 subjects wearing and ECG sensor, respiratory sensor, and the Empatica E4. The analysis in Arce and Gehringer (2021) will be completed for the datasets from Thayer and Stevens (2021), where the data is used to train four regression models, four multi-class models, and two binary models and standard performance metrics are calculated.

Thayer and Stevens (2021) conducted two studies (n=73, N=73) in which participants first completed a series attentional control and working memory tasks followed by either a brief interaction with a dog or a control task followed by another round of attention and memory tasks. Thus, there are three phases (pre-intervention, intervention, and post-intervention) and two between-subjects groups for the intervention (animal interaction, control task). Participants were randomly assigned to the animal or control intervention. Repeatedly over the experimental session, participants self-reported mood, anxiety, and stress levels. Continuously throughout the session, we recorded electrodermal activity, heart activity, temperature, and movement with an Empatica E4.

Randomization In the original study, participants were randomly assigned to condition. In the current analysis, the program will randomly select rows of data for the training vs. testing sets.

Sampling Plan

Existing data **Registration prior to analysis of the data**. As of the date of submission, the data exist and you have accessed it, though no analysis has been conducted related to the research plan (including calculation of summary statistics). A common situation for this scenario when a large dataset exists that is used for many different studies over time, or when a data set is randomly split into a sample for exploratory analyses, and the other section of data is reserved for later confirmatory data analysis.

Explanation of existing data

At this stage, we have compiled the data from multiple participants into a single data file using automated scripts. But no data feature extraction has begun, no machine learning training has been performed, and no outcome measures have been generated from a trained machine learning model.

Data collection procedures

There is no active data collection for this stage of the project, as it uses existing data. All future data will be compiled by Microsoft.ML and the Biosensor Framework that we are using for the machine learning training.

Sample size

There are 146 participants (73 in each of two studies) that experienced the experimental protocol in Thayer and Stevens (2021), with sessions lasting between 30 and 40 minutes. All subjects will be analyzed as part of this study.

Sample size rationale

We will use all samples collected in the original data.

Stopping rule

N/A

Variables

Manipulated variables

In the original Thayer & Stevens (2021) study, the manipulated variable was the interaction type: active (HAI condition) or passive (control condition).

The current analysis has no manipulated variables.

Measured variables

The Empatica E4 is a wrist worn biosensor that outputs four continuous data streams: electrodermal activity, photoplethysmography, accelerometry (x, y, z), and temperature. The electrodermal activity is measured by sensing the resistance of the skin on the ventral side of the wrist. The photoplethysmography is an analogue to an electrocardiogram (ECG) and is measured by sensing the volumetric change in blood in peripheral circulation on the top of the wrist, which also acts as an implicit measure of heart activity. The accelerometry is captured using a 3-axis accelerometer on the back of the wrist which senses the change in acceleration at a

point. The temperature data is captured using a thermopile on the top of the wrist and measures temperature in Celsius.

We are classifying five different states: cognitive load before intervention, active intervention, passive intervention, and cognitive load after intervention.

Cognitive load is a state where the subject is performing cognitive tasks that require work to complete, thereby increasing the amount of attention and energy expended during the task (pre-intervention and post-intervention phases). We treat both cognitive load states as onset conditions where we expect markers of stress to correspond to psychological state.

Active intervention is a state where there is a presupposed positive stimulus, in this case a dog, that is allowed to interact with the subject to reduce the effects of cognitive load. This intervention is randomly selected for the subjects.

Passive intervention is a state where there is no presupposed positive stimulus, in this case no dog, that interacts with the subject. This intervention is also randomly selected for the subjects.

Indices The features derived from the raw data consist of statistical and frequency response features. The Biosensor Framework converts the raw data streams into 119 feature vectors. Each axis of the accelerometry produces 4 features, with the 3-axis raw data being used to produce 4 more. The electrodermal activity produces 15 features. The photoplethysmography produces 18 features. Finally, the temperature produces 6 features.

These features are generated in 5 second windows. So these 119 features will be calculated for each 5 second chunk and the final collection of vectors is the condensed dataset for the subject. These features consist of the maximum and minimum of the raw data, the standard deviation of the raw data, the absolute value of the area under the curve for the integral of the signal, the peak frequency of the signal using a Fast Fourier transform, the root mean square of the signal, the dynamic range of the signal, and the slope of the signal.

To process the photoplethymography signal, the first derivative is estimated using the finite difference method and the local maxima are found. Using the maxima and the sampling rate (64 Hz), the heart rate, RR intervals, NN50, and pNN50

are calculated. Additionally, the raw data is used to calculate the frequency band energies, which are used to calculate the frequency ratio, normalized to find the mean, and used to find the signal frequency power.

To process the electrodermal activity, the phasic (SCR) and tonic (SCL) components are found by removing the trend using the Tarvainen method and finding the residual by subtracting the SCR from the raw signal. The raw signal is also used to find the magnitude, duration, and amplitude of the various peaks within the signal using the Healey method.

Analysis Plan

Statistical models

The raw data will be split using a 70/30 split, so 70% of the data will be used to train and 30% will be used to evaluate the performance of the models. The evaluation dataset will be used to generate our summary statistics on model performance, which allows the model to be evaluated using data that it has not seen. During the evaluation phase, the model will not be training (i.e. updating weights), so it will not get better or worse as it evaluates.

The models built into Microsoft.ML will be used. One of the constraints with Microsoft.ML at the time of creating the Biosensor Framework was that many models were still in development and some had bugs that made them inoperable. Due to this, the Biosensor Framework excluded them. This study will reevaluate the models used and add in models that are working. The current benchmarking models in the Biosensor Framework are explained below.

The multi-class models consist of: Averaged Perceptron, Fast Forest, Fast Tree, LBFGS Logistic Regression, LBFGS Maximum Entropy, Light GBM, Linear SVM, SDCA Maximum Entropy, SGD Calibrated, and K Means. Each multi-class model will be evaluated using their log loss, log loss reduction, macro accuracy, micro accuracy, and class precision, recall, and loss.

The regression models consist of: Fast Forest, Fast Tree, Fast Tree Tweedie, LBFGS Poisson Regression, and Light GBM. Each regression model will be evaluated using mean absolute error, mean squared error, root mean square error, and r-squared.

The binary models consist of: Fast Forest and Fast Tree. Each binary model will

be evaluated using accuracy, area under the precision-recall curve, area under ROC curve, F1 score, negative precision and recall, and positive precision and recall.

The feature importance will be evaluated using permutation feature importance.

For all available regression models in the ML.NET API: https://docs.microsoft.com/en-us/dotnet/api/microsoft.ml.regressioncatalog.regressiontrainers?view=ml-dotnet

For all available multi-class models in the ML.NET API: https://docs.microsoft.com/en-us/dotnet/api/microsoft.ml.multiclassclassificationcatalog.multiclassclassificationtrainers?view=ml-dotnet

For all available binary models in the ML.NET API: https://docs.microsoft.com/en-us/dotnet/api/microsoft.ml.binaryclassificationcatalog.binaryclassificationtrainers?view=ml-dotnet

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Transformations

Once the raw data has been converted to feature vectors, they are used as-is for training, testing, and validation.

Inference criteria

We will use Bayes factors for comparing which algorithms have the highest accuracy. We will use the thresholds from Wagenmakers et al. (2018), with Bayes factors $>$ 3 indicating moderate evidence for the highest performing model compared to the next model.

Walker, will we get accuracy/importance measures with multiple repetitions for each algorithm and feature (if that makes sense)? That is, will we be able to compare a distribution of accuracies for one algorithm to a distribution for another algorithm? Or will we just get a single accuracy for each algorithm? Same for feature importance.

Data exclusion

No outliers are expected.

Missing data

Any missing or incomplete data will be excluded from the dataset in the processing step.

Exploratory analyses (optional)	Enter your response here.

Other

Other (Optional)	Enter your response here.

References